南方硅谷 SSV6358
基于 BlueZ 的 BLE 使用说明

简介

文档主要介绍 6358 系列芯片 ble 功能的使用介绍。

发布说明：

版本	日期	作者	描述
V1.0	20230306	icommsemi	初版

目录

1. 驱动编译和加载

1.1 内核蓝牙协议栈配置

使用 Bluez 协议栈需要 Linux kernel 的 bluetooth feature 支持，需要在 kernel_menuconfig 或 menuconfig 选上对应的配置,不同芯片提供的内核配置会一些差异，大体配置相同，以上提供参考：

1） 配置使能内核部分蓝牙功能：make kernel_menuconfig

```
[*]   Networking support --->
  <*>   Bluetooth subsystem support --->
      [*]   Bluetooth Low Energy (LE) features
```

2） 配置蓝牙协议支持 UART 接口设备: make kernel_menuconfig

```
[*]   Networking support --->
  <*>   Bluetooth subsystem support --->
          Bluetooth device drivers --->
        <*> HCI UART driver
        [*]   UART(H4) protocol support
```

3） 建议验证及使用驱动 BLE feature 的内核版本及 Bluez 的版本

Linux kernel >= v4.4

Bluez >= v5.50

1.2 基于 USB 或 SDIO 接口

1）驱动编译选项修改 config.mak

```
## BLE HCI usage (注意与NIMBLE的使用为2选1)
CONFIG_BLE ?= y
ccflags-$(CONFIG_BLE) += -DCONFIG_BLE

# BLE HCI BUS 取值：
# CONFIG_BLE_HCI_OVER_UART:0
# CONFIG_BLE_HCI_OVER_HWIF:1  USB or SDIO, base on HWIF
ccflags-y += -DCONFIG_BLE_HCI_BUS=1
```

2）配置文件修改 firmware image 路径及加载 bin 文件名称，使用 ble 功能需使用名称为 ssv6x5x-sw_ble.bin 的文件

```
################################################################################
# Firmware setting
#
# Priority.1 insmod parameter "cfgfirmwarepath" (insmod的時候帶入)
# Priority.2 firmware_path (在下面設定)
# Priority.3 default firmware (前兩個都沒設定會load firmware header)
#
################################################################################
#firmware_path = /home/xxx/xxxx/image/
firmware_path = /home/icomm/Downloads/6358/ssv6x5x/image/
################################################################################
# Firmware name
#
# 若有設定firmware path的話才有用, 內部預設firmware name使用"ssv6x5x-sw.bin"
#
################################################################################
#firmware_name = ssv6x5x-sw.bin
firmware_name = ssv6x5x-sw_ble.bin
```

3）执行 load.sh 加载驱动后通过 kernel 中 log 信息确认 BLE HCI 功能，USB 或 SDIO 接口

```
[62944.408419] ble_hci_init
[62944.408420] cfg.ble_dtm 0
[62944.408425] hci device name:SSV6XXX_USB
[62944.408593] \x1b[32mssv_ble_hci_open():78 \x1b[0m
[62944.408595] Config BLE HCI over HWIF
[62944.409102] hci device id:1
```

4）测试 HCI 应用接口：

以下使用 USB 接口接入基于 ubuntu 系统，该说明使用基于 Ubuntu 22.04, BlueZ 5.64 版本测试

- **确认加载 HCI 设备成功**

  sudo hciconfig

```
icomm@icomm-virtual-machine:~/Downloads/6358/ssv6x5x$ hciconfig
hci1:   Type: Primary  Bus: USB
        BD Address: 14:B2:E5:FF:AA:E9  ACL MTU: 27:8  SCO MTU: 0:0
        DOWN
        RX bytes:0 acl:0 sco:0 events:14 errors:0
        TX bytes:0 acl:0 sco:0 commands:14 errors:0
```

- **启动 HCI 设备开始工作**

  sudo hciconfig hci1 up

```
icomm@icomm-virtual-machine:~/Downloads/6358/ssv6x5x$ sudo hciconfig hci1 up
icomm@icomm-virtual-machine:~/Downloads/6358/ssv6x5x$ hciconfig
hci1:   Type: Primary  Bus: USB
        BD Address: 14:B2:E5:FF:AA:E9  ACL MTU: 27:8  SCO MTU: 0:0
        UP RUNNING
        RX bytes:0 acl:0 sco:0 events:30 errors:0
        TX bytes:0 acl:0 sco:0 commands:30 errors:0
```

1.3 基于 UART 接口

1）驱动编译选项 config.mak,修改

```
## BLE HCI usage (注意与NIMBLE的使用为2选1)
CONFIG_BLE ?= y
ccflags-$(CONFIG_BLE) += -DCONFIG_BLE

# BLE HCI BUS 取值：
# CONFIG_BLE_HCI_OVER_UART:0
# CONFIG_BLE_HCI_OVER_HWIF:1  USB or SDIO, base on HWIF
ccflags-y += -DCONFIG_BLE_HCI_BUS=0
```

2）配置文件修改 firmware image 路径及加载 bin 文件名称，使用 ble 功能需使用名称为 ssv6x5x-sw_ble.bin 的文件

```
##############################################################################
# Firmware setting
#
# Priority.1 insmod parameter "cfgfirmwarepath" (insmod的時候帶入)
# Priority.2 firmware_path (在下面設定)
# Priority.3 default firmware (前兩個都沒設定會load firmware header)
#
##############################################################################
#firmware_path = /home/xxx/xxxx/image/
firmware_path = /home/icomm/Downloads/6358/ssv6x5x/image/
##############################################################################
# Firmware name
#
# 若有設定firmware path的話才有用, 內部預設firmware name使用"ssv6x5x-sw.bin"
#
##############################################################################
#firmware_name = ssv6x5x-sw.bin
firmware_name = ssv6x5x-sw_ble.bin
```

3）执行 load.sh 加载驱动后通过 kernel 中 log 信息确认 BLE HCI UART 功能

```
[57588.424058] ieee80211 phy4: New interface create p2p0
[57588.424071] SSV host driver version 1400, Build Data 2023-02-22
[57588.432421] SSV firmware version 10947
[57588.432425] ble_hci_init
[57588.432426] Config BLE HCI over UART
[57588.432472] ~~~~~~~~~INIT SSV CTL GENERIC NETLINK MODULE
[57588.432479] ~~~~~~~~~INIT SSV WPAS CTL GENERIC NETLINK MODULE
[57588.432485] SSV6X5X of iComm-semi
[57588.432485] ssv_custom_modify_rf_conf_table
```

4）硬件接口：
PIN30- GPIO 36 - UART2-RXD
PIN31- GPIO 37 - UART2-TXD

5）串口参数：115200-8N1

6）测试 HCI uart 应用接口：
以下例子使用 USB 接口，串口 txrx 通过电平转换接入，基于 ubuntu 系统，检测到串口设备为/dev/ttyUSB0，该说明使用基于 Ubuntu 22.04, BlueZ 5.64 版本测试

- **挂载 HCI 设备，波特率为 115200**
  sudo hciattach -s 115200 /dev/ttyUSB0 any 115200 noflow nosleep

- **查看加载的 HCI 设备号**
  sudo hciconfig

```
icomm@icomm-virtual-machine:~/Downloads/6358/ssv6x5x$ hciconfig
hci1:   Type: Primary  Bus: UART
        BD Address: 00:00:00:00:00:00  ACL MTU: 0:0  SCO MTU: 0:0
        DOWN
        RX bytes:0 acl:0 sco:0 events:0 errors:0
        TX bytes:0 acl:0 sco:0 commands:0 errors:0
```

- **启动 HCI 设备开始工作**
  sudo hciconfig hci1 up

```
icomm@icomm-virtual-machine:~/Downloads/6358/ssv6x5x$ sudo hciconfig hci1 up
icomm@icomm-virtual-machine:~/Downloads/6358/ssv6x5x$ hciconfig
hci1:   Type: Primary  Bus: UART
        BD Address: 14:B2:E5:FF:AA:E9  ACL MTU: 27:8  SCO MTU: 0:0
        UP RUNNING
        RX bytes:200 acl:0 sco:0 events:13 errors:0
        TX bytes:68 acl:0 sco:0 commands:13 errors:0
```

2. 配置文件中 BLE 设置说明

2.1 BLE 的 MAC 地址设置

目前支持使用的是 public address

```
#########################################################################
# MAC address setting
#
# 設定裝置的MAC address,
# Priority.1 hw_mac (在下面設定)
# Priority.2 efuse存在的MAC address
# Priority.3 default MAC (前兩個都沒設定會用驅動寫死的MAC address)
#
#########################################################################
hw_mac = 00:23:45:67:89:20
```

按优先级支持 3 种方式配置 MAC 地址，默认选择优先级最高（priority 1）

如图所示，配置 hw_mac = 00:23:45:67:89:20，则此时 BLE MAC 地址为 00:23:45:67:89:21

```
icomm@icomm-virtual-machine:~/Downloads/6358/ssv6x5x$ hciconfig
hci1:   Type: Primary  Bus: USB
        BD Address: 00:23:45:67:89:21  ACL MTU: 27:8  SCO MTU: 0:0
        UP RUNNING
        RX bytes:0 acl:28 sco:0 events:732 errors:0
        TX bytes:0 acl:25 sco:0 commands:70 errors:0
```

若未配置 hw_mac,则取 efuse 存在的 MAC address，否则使用默认值

2.2 进入 BLE DTM 测试模式

测试模式使能需在配置文件中使能 ble_dtm = 1（去掉#）,此时重启加载驱动后，进入 BLE DTM 测试模式，其他功能暂不可用；

```
##################################################
# BLE Setting
# if ble_replace_scan_win and ble_replace_scan_interval both are not 0
# only apply when stack set full scan to link layer
##################################################
#ble_replace_scan_win = 0x30
#ble_replace_scan_interval = 0x0140
#ble_dtm = 1
```

- 硬件接口：
  PIN30- GPIO 36 - UART2-RXD
  PIN31- GPIO 37 - UART2-TXD
- 串口参数：115200-8N1

3. Bluez 测试 BLE 基本功能说明

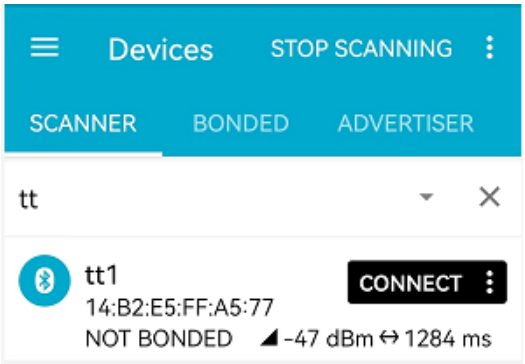
3.1 作为 peripheral 测试广播连接

以下使用 bluez 自带的 bluetoothctl 工具来测试（版本：5.64）

进入 bluetoothctl 命令交互界面：

- 在菜单广播设置中，配置广播名称
  [Bluetooth]# menu advertise
  [Bluetooth]# name tt1
  [Bluetooth]# back
- 返回主菜单页面后打开广播
  [Bluetooth]# advertise on

此时打开手机 BLE 测试 APP(如 nrfconnect)，可以扫描到设备



默认广播的间隔为 1280ms

√如何修改广播间隔

根据不同平台关于 bluetooth 功能配置的不同，这里介绍 3 种方式（在 5.64 版本测试，其他版本需自行检查是否支持特性）

1）配置 /etc/Bluetooth/main.conf 中的广播参数 MinAdvertisementInterval/ MaxAdvertisementInterval，应用初始化加载 main.conf 的默认参数

2）应用启动带-E 参数启动实验性特性，通过 bluetoothctl 中 interval 命令可以修改

3）修改/sys/kernel/debug/bluetooth/hcix 中的参数值 adv_min_interval/ adv_max_interval（需内核支持 debugfs 的 bluetooth 功能）

关于扫描/连接参数的设置方法可以参考上述介绍。

- 测试 GATT server 连接与读写 characteristic

从主菜单进入 gatt 配置菜单，新增用于测试的 service 和 characteristics

[Bluetooth]# menu gatt

[Bluetooth]# register-service ff01

提示是否 primary：yes

[Bluetooth]# register-characteristics ff02 read,write

提示输入 Value 值：（多字节需带空格，数值为十进制输入）

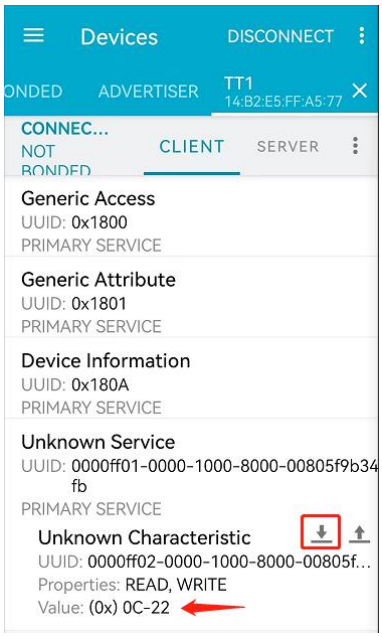
Enter value 12 34

[Bluetooth]# register-application

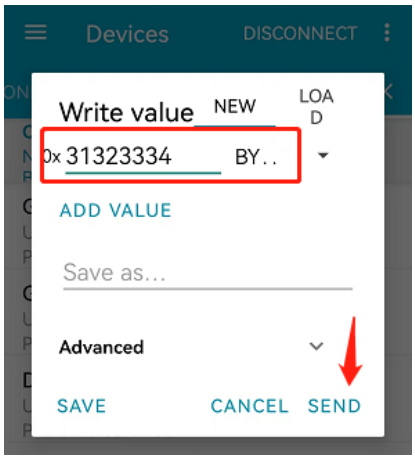
```
[bluetooth]# register-service ff01
[□□NEW□□] Primary Service (Handle 0x0000)
        /org/bluez/app/service0
        ff01
[/org/bluez/app/service0] Primary (yes/no): yes
[bluetooth]# register-characteristic ff02 read,write
[□□NEW□□] Characteristic (Handle 0x0000)
        /org/bluez/app/service0/chrc0
        ff02
[/org/bluez/app/service0/chrc0] Enter value: 12 34
[bluetooth]# register-application
[□□CHG□□] Primary Service (Handle 0x0015)
        /org/bluez/app/service0
        ff01
[□□CHG□□] Characteristic (Handle 0x0017)
        /org/bluez/app/service0/chrc0
        ff02
```

此时在进入广播后，打开手机 APP-nrfconnect，扫描到设备后点击[CONNECT],等待连接成功；

如图所示：点击  read value 等于(0x)0C -22



点击write value：0x31323334 （BYTE ARRAY），



设备侧显示收到数据 3132334

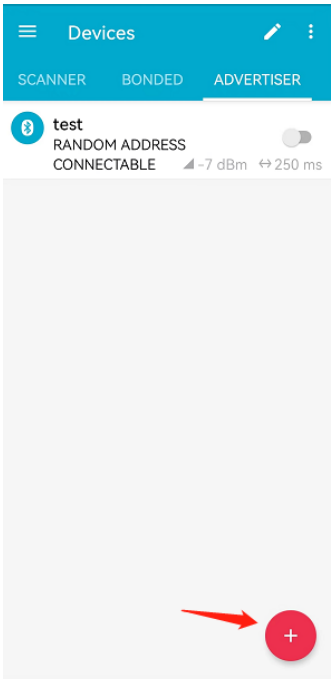
```
[/org/bluez/app/service0/chrc0 ((null))] ReadValue: 30:A1:FA:37:D8:07 offset 0 link LE
[/org/bluez/app/service0/chrc0 ((null))] WriteValue: 30:A1:FA:37:D8:07 offset 0 link LE
  31 32 33 34                                      1234
```

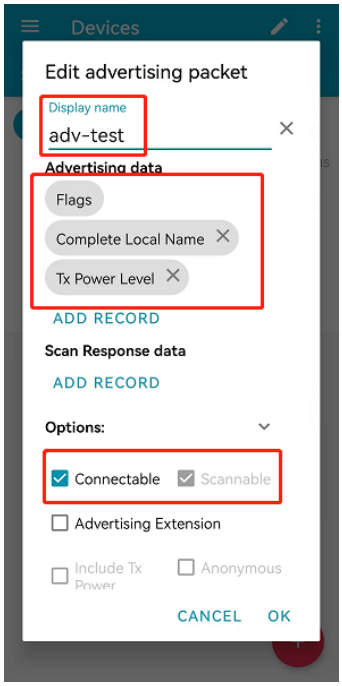
3.2 作为 central 测试扫描连接

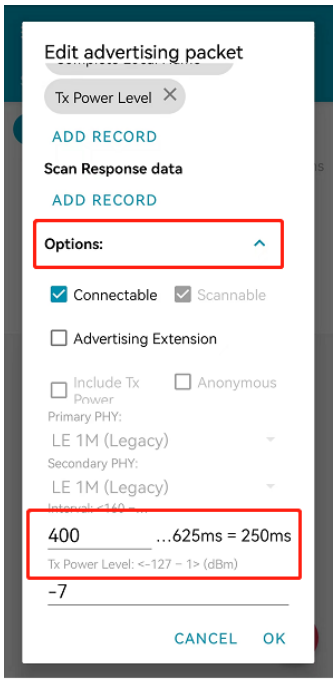
以下使用 bluez 自带的 bluetoothctl 工具来测试（版本：5.64）

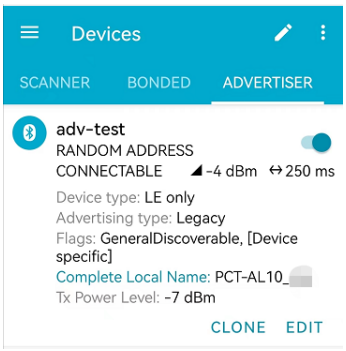
- 手机 app-nrfconnect 设置发起广播，

点击创建 advertiser，设置显示 name 及广播数据，选项勾选 conntectable/scannable，点击 Options，设置广播间隔后打开广播，如下图所示：









- 进入 Bluetoothctl 命令行主菜单界面发起扫描设备

[Bluetooth]# scan on

```
[bluetooth]# scan on
Discovery started
[CHG] Controller 00:23:45:67:89:21 Discovering: yes
[NEW] Device 4E:CA:36:D8:C2:0C 4E-CA-36-D8-C2-0C
[NEW] Device 9C:C1:2D:7C:B9:EE midea
[NEW] Device 41:D5:9A:69:64:FF 41-D5-9A-69-64-FF
[NEW] Device 72:02:C2:DE:3A:E8 72-02-C2-DE-3A-E8
[NEW] Device 67:49:48:5E:B2:08 67-49-48-5E-B2-08
[NEW] Device 54:A4:93:68:E0:D9 54-A4-93-68-E0-D9
[NEW] Device DA:00:83:98:5E:E6 DA-00-83-98-5E-E6
[NEW] Device 69:94:77:14:A3:9F PCT-AL10_
[NEW] Device 71:5D:88:C1:A2:39 71-5D-88-C1-A2-39
[NEW] Device 7E:2E:C3:A0:41:1C 7E-2E-C3-A0-41-1C
[NEW] Device 6A:77:06:81:1D:80 6A-77-06-81-1D-80
[CHG] Device 67:49:48:5E:B2:08 RSSI: -82
```

- 连接设备，需要 scan 到设备才能进行连接操作，否则返回连接失败

[Bluetooth]# connect 69:94:77:14:A3:9F

- 连接设备成功后，可以同样操作 read/write value，参考上一节操作
- 断开设备连接

```
[PCT-AL10_    ]# disconnect
Attempting to disconnect from 69:94:77:14:A3:9F
Successful disconnected
```